

2023



Nestlé Efficiently Scales to 2.8 Million Connected Devices with AWS IoT

Learn how Nestlé developed a global framework for Internet of Things products using AWS IoT Core.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Other](#)



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2.8M devices connected

Reduced IoT solution development



to a centralized IoT Platform

from months to weeks

Supports connected consumer devices

across 97 countries

Created a repeatable and scalable

technology model to accelerate innovation

Unlocked new functionality

to improve customer experience

Overview

Consumer packaged goods (CPG) giant [Nestlé](#) wanted to accelerate the development of Internet of Things (IoT) solutions across its global business units, including coffee brand [Nespresso](#) and pet care brand [Nestlé Purina Petcare](#) (Purina). Rather than have each business unit build its own IoT offering, Nestlé set out to build a single centralized framework, the IoT Platform, that could be adopted by numerous brands. “Our main goal is to reuse common components across solutions and accelerate the creation of new customer experiences,” says Víctor García, senior product manager for the consumer IoT Platform at Nestlé.

It’s safe to say that there has never been a project quite like this in Nestlé’s storied history, which stretches across well over 150 years. Yet, the IT team was determined to break down walls and deliver a first-of-a-kind initiative. Nestlé chose to build the IoT Platform on Amazon Web Services (AWS) to unlock scale and take advantage of serverless architecture. “We have no doubt that we picked the right provider when we chose AWS,” says Adrian Matellanes, global head of digital experience technologies at Nestlé.

Opportunity | Using AWS IoT Core to Accelerate Innovation for Nestlé

As one of the largest CPG corporations in the world, Nestlé is home to many household names. To accelerate time to market for IoT innovation across its global businesses, Nestlé needed a unified and secure solution framework that would liberate teams to focus on the customer experience. “Some of our brands wouldn’t have been able to afford or develop IoT solutions in house if we didn’t have a common solution ready for them to use,” says Matellanes. Nestlé used [AWS IoT Core](#), a service that offers cloud connectivity for billions of IoT devices across the globe without managing infrastructure, to build its central offering.

With the IoT Platform available, Nestlé business units have been freed to innovate and focus their resources on building differentiated products. At Purina, for example, teams used the IoT Platform to bring essential and often complex IoT functionalities—such as device-to-cloud connectivity, data ingestion, security certificate management, and software update management—to its newly developed Smart Litterbox offering, which uses artificial intelligence (AI) to alert people when their cats may need to see the veterinarian. “The IoT Platform was a significant catalyst for us to get buy-ins from business stakeholders and to off-load complex IoT workflows,” says Ann Scharenberg, director of architecture and services for new business models at Purina US. “We’re using it to build a number of new products.”

Solution | Global Head of Digital Experience Technologies, Nestlé

From the moment the IoT Platform initiative started in 2016, Nestlé IT leaders were set on using serverless architecture. “The single most important differentiator for AWS was its serverless computing service,” says García. “Just as we wanted to take common components out of our IT solutions so teams could focus on business value, we wanted to use AWS to reduce the load of managing infrastructure.” Nestlé uses [AWS Lambda](#), a serverless, event-driven compute service, so its teams can focus on the consumer experience.

Seven years is a long time, and Nestlé has been busy. After its success with its first proof of concept, which was for a [Nescafé](#) product, the corporation has incrementally expanded its IoT operations throughout its global business units. Today, over 2.8 million devices in 97 countries are connected through the IoT Platform, which has ingested 2.1 billion data points in that time. “When so many of our companies grew and extended the technology spoke for itself,” says Matellanes. And as the enterprise gains more experience building IoT solutions, development times have shrunk from months to weeks.

Purina is the most recent Nestlé company to bring a connected device to consumers using the IoT Platform, which is powered by AWS IoT Core. The business worked closely with [AWS Professional Services](#), a team that assists organizations with enterprise cloud computing initiatives, to realize its AI-powered Smart Litterbox project. “Working with AWS Professional Services, we mapped out a service-based architecture to meet many requirements for customers with multiple devices, all with the IoT Platform continuing to drive the consistency and quality that we’ve seen for years,” says Scharenberg. Purina also used [Amazon SageMaker](#), a service to build, train, and deploy machine learning models, to develop the AI for the Smart Litterbox. Once the device shipped to retail stores and Amazon in May of 2023, the team started seeing more detailed data than it’s ever received from any product.

For Purina, IoT devices are about more than data—they’re about building a relationship with consumers. The Smart Litterbox is a case in point. Purina engineered the product to provide pet owners with detailed information and proactive health predictions throughout the full pet life cycle. By helping people care for their beloved pets, Purina is building trust that will elevate its brand identity. “We think this product will change our consumer relationships in a way that wouldn’t be possible with our other products,” says Scharenberg. “From our perspective, the potential is huge, and it will only grow as we add more devices into our IoT portfolio.”

Outcome | Advancing IoT Innovation around the World on AWS

Armed with the IoT Platform, Nestlé teams are ready to pick up the pace of innovation. In the coming years, business units, including Purina, are set to roll out numerous IoT products. “The constant evolution of AWS offerings keeps us on our toes,” says Matellanes. “We have absolute peace of mind with our systems on AWS.”

Due to the sheer scale and complexity of a global enterprise like Nestlé, consistency has been key to the success of IoT initiatives. “The story of this solution’s continuous evolution over 7 years is one of constant standardization and improvement,” says García. “And as we worked, AWS kept up a constant pace of innovation, delivering new specialized services that we used to accelerate our platform’s value proposition.”

About Nestlé

Nestlé is a global CPG company headquartered in Switzerland. It has several business units, including Nespresso, Stouffer’s, Nestlé Purina Healthcare, and many more, to fit virtually any aspect of life under its motto: Good Food, Good Life.

AWS Services Used

AWS IoT Core

AWS IoT Core lets you connect billions of IoT devices and route trillions of messages to AWS services without managing infrastructure.

[Learn more »](#)



AWS Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

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Amazon SageMaker

Build, train, and deploy machine learning (ML) models for any use case with fully managed infrastructure, tools, and workflows.

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